

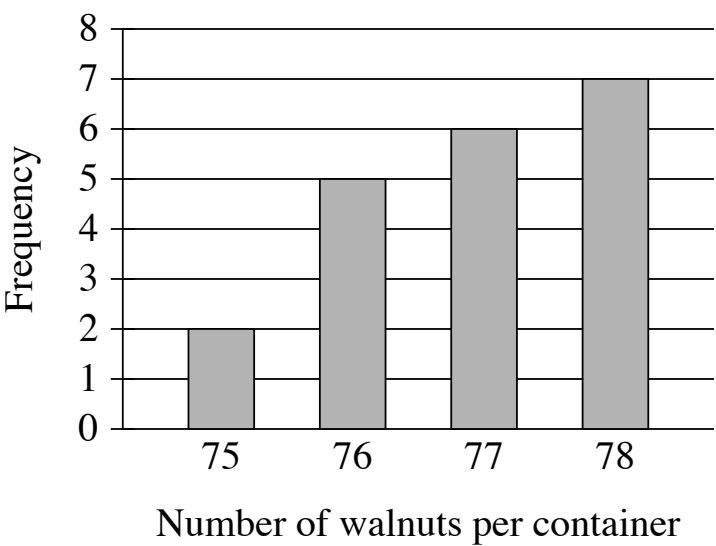
Question ID fe6a49d6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: fe6a49d6

PSAT2 M 2.1

The bar graph shows the distribution of the number of walnuts per container for **20** containers at a grocery store.



How many of these containers of walnuts contain exactly **78** walnuts?

- A. **2**
- B. **7**
- C. **20**
- D. **78**

Question ID 771ee744

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 771ee744

PSAT2 M 2.2

There are **320** marbles in a container. Of these marbles, **10%** are red. How many marbles in the container are red?

- A. **32**
- B. **288**
- C. **320**
- D. **352**

Question ID fe4c1c9e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: fe4c1c9e

PSAT2 M 2.3

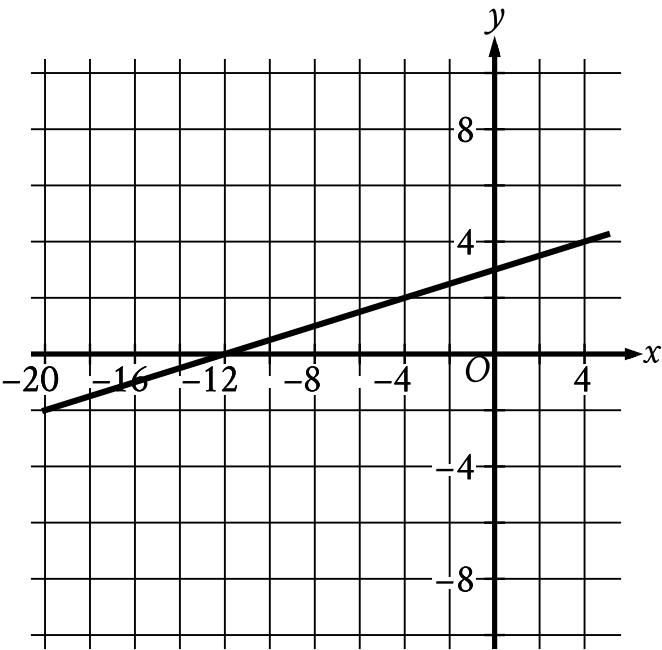
A mechanical device in a workshop produces items at a constant rate of **60** items per hour. At this rate, how many items will the mechanical device produce in **3** hours?

Question ID c10ad793

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: c10ad793

PSAT2 M 2.4



The graph of the linear function f is shown, where $y = f(x)$. What is the x-intercept of the graph of f ?

- A. $(-12, 0)$
- B. $(0, 0)$
- C. $(\frac{1}{4}, 0)$
- D. $(12, 0)$

Question ID c6a26e14

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: c6a26e14

PSAT2 M 2.5

$|x + 45| = 48$ What is the positive solution to the given equation?

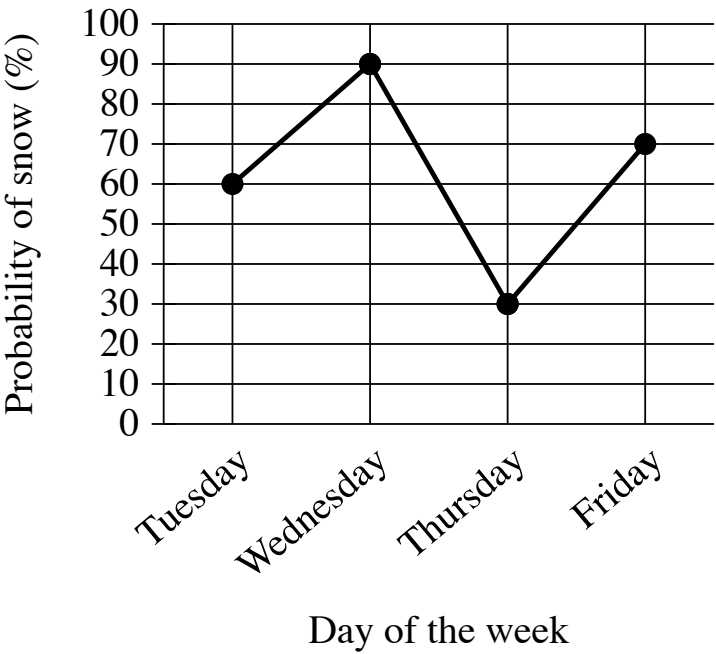
- A. 3
- B. 48
- C. 93
- D. 96

Question ID 13f67ddc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 13f67ddc

PSAT2 M 2.6



The line graph shows the probability of snow, as a percent, at a certain location for each day during a four-day period. According to the line graph, for which day during this four-day period is the probability of snow **30%**?

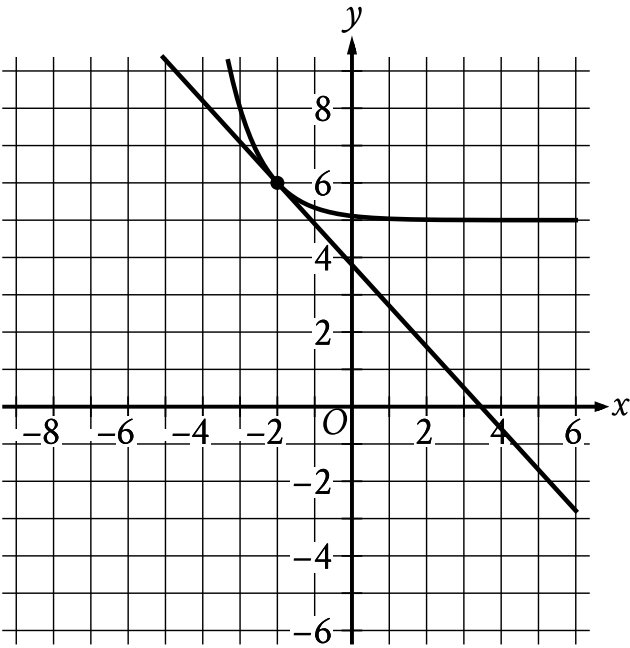
- A. Tuesday
- B. Wednesday
- C. Thursday
- D. Friday

Question ID 5c7d5744

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 5c7d5744

PSAT2 M 2.7



The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

- A. $(6, 0)$
- B. $(-2, 6)$
- C. $(0, -2)$
- D. $(0, 0)$

Question ID e0177f5f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: e0177f5f

PSAT2 M 2.8

$x = 4$ $y = 5 - x$ The solution to the given system of equations is (x, y) . What is the value of y ?

- A. 1
- B. 4
- C. 5
- D. 9

Question ID e006209c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: e006209c

PSAT2 M 2.9

A geologist needs to collect at least **67** samples of lava from a volcano. If the geologist has already collected **63** samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

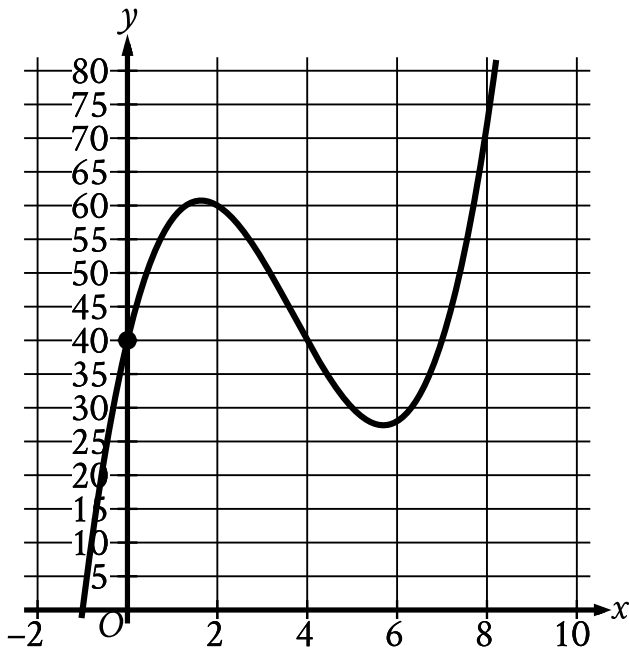
- A. **130**
- B. **63**
- C. **4**
- D. **0**

Question ID 26f5269a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 26f5269a

PSAT2 M 2.10



The y -intercept of the graph shown is (x, y) . What is the value of y ?

Question ID f2b63f49

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: f2b63f49

PSAT2 M 2.11

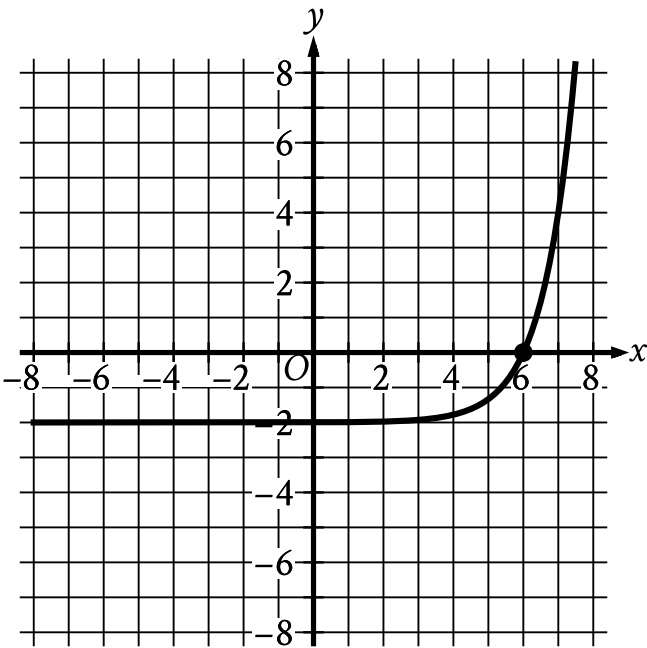
$8x - 7x + 130 = 260$ What value of x is the solution to the given equation?

Question ID 2b6c12eb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 2b6c12eb

PSAT2 M 2.12



What is the x-coordinate of the x-intercept of the graph shown?

Question ID 1d18794b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 1d18794b

PSAT2 M 2.13

A contract for a certain service requires a onetime activation cost of **\$35** and a monthly cost of **\$23**. Which equation represents this situation, where *c* is the total cost, in dollars, of this service contract for *t* months?

A. $c = \frac{t}{23} + 35$

B. $c = \frac{t}{35} + 23$

C. $c = 23t + 35$

D. $c = 35t + 23$

Question ID 4e97f862

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 4e97f862

PSAT2 M 2.14

The function f is defined by $f(x) = 3x - 8$. What is the value of $f(7)$?

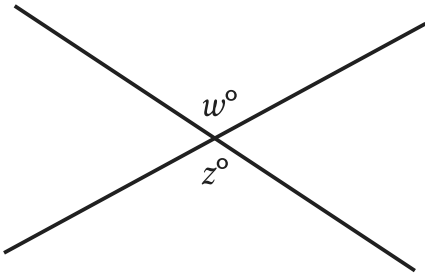
- A. 29
- B. 13
- C. −5
- D. −29

Question ID 64d1f49f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 64d1f49f

PSAT2 M 2.15



Note: Figure not drawn to scale.
In the figure, two lines intersect at a point. If $w = 136$, what is the value of z ?

- A. 36
- B. 44
- C. 68
- D. 136

Question ID 4efea6a3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 4efea6a3

PSAT2 M 2.16

The area of a rectangle is **57** square inches. The length of the longest side of the rectangle is **19** inches. What is the length, in inches, of the shortest side of this rectangle?

Question ID 02489d55

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 02489d55

PSAT2 M 2.17

Which expression is equivalent to $19(x^2 - 7)$?

- A. $19x^2 - 133$
- B. $19x^2 - 26$
- C. $19x^2 - 7$
- D. $19x^2 + 12$

Question ID bd4d0e0c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: bd4d0e0c

PSAT2 M 2.18

The function f is defined by $f(x) = 10x^2 - 32x - 152$. What is the value of $f(0)$?

- A. -152
- B. -32
- C. 0
- D. 10

Question ID 0bebc08c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 0bebc08c

PSAT2 M 2.19

$x = 3$ $y = (15 - x)^2$ A solution to the given system of equations is (x, y) . What is the value of xy ?

- A. 432
- B. 54
- C. 45
- D. 18

Question ID 301faf80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 301faf80

PSAT2 M 2.20

The product of two positive integers is ~~4~~**62**. If the first integer is **5** greater than twice the second integer, what is the smaller of the two integers?

Question ID df747160

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: df747160

PSAT2 M 2.21

Which expression is equivalent to $17(x^2 - 100y^2)$?

- A. $17(x - 2y)(x - 50y)$
- B. $17(x - 2y)(x + 50y)$
- C. $17(x - 10y)(x - 10y)$
- D. $17(x - 10y)(x + 10y)$

Question ID 797a81fb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 797a81fb

PSAT2 M 2.22

$-12x + 14y = 36$ $-6x + 7y = -18$ How many solutions does the given system of equations have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero